University of Debrecen, Faculty of Informatics

Logic in computer science

First-order logic

(predicate logic)

semantic rules, evaluation

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First-order interpretation

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first-order language. The pair $\langle U, \varrho \rangle$ is the *interpretation* of $L^{(1)}$ if

- U is a nonempty set of objects; and
- ϱ is a mapping such that $Dom(\varrho) = Con$ and
  - if $a \in \mathcal{F}(0)$, akkor $\varrho(a) \in U$;
  - if $f \in \mathcal{F}(n) \quad (n > 0)$, then

$$\varrho(f) \in U^{U^n} \text{ and so } \varrho(f) : U^n \to U;$$

  - if $p \in \mathcal{P}(0)$, akkor $\varrho(p) \in \{0, 1\}$;
  - if $P \in \mathcal{P}(n) \quad (n > 0)$, then

$$\varrho(P) \in \{0, 1\}^{U^n} \text{ and so } \varrho(P) : U^n \to \{0, 1\}.$$

Assignment

Variable assignment

Let $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first-order language and $\langle U, \varrho \rangle$ be an interpretation of $\mathcal{L}^{(1)}$. The v mapping is an *assignment* relying on the interpretation $\langle U, \varrho \rangle$ if:

$$v : Var \to U$$

Modified assignment

Let v be an assignment relying on the interpretation $\langle U, \varrho \rangle$, $x \in Var$ be a variable and $u \in U$ be an object. If it is true for all $y \in Var$ that

$$v[x : u](y) = \begin{cases} u, & if \quad y = x; \\ v(y), & \text{otherwise.} \end{cases}$$

then $v[x : u]$ is a *modified assignment*. The $v[x : u]$ function is an *x-variant* of v.

First-order logic – evaluation of terms

Let $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first-order language, $\langle U, \varrho \rangle$ be one of its interpretations and v be an assignment relying on $\langle U, \varrho \rangle$.

- If $a \in \mathcal{F}(0)$, then $|\, a \,|_v^{\langle U, \varrho \rangle} = \varrho(a)$.
- If $x \in Var$, then $|\, x \,|_v^{\langle U, \varrho \rangle} = v(x)$.
- If $f \in \mathcal{F}(n)$ for some $n > 0$, then

$$|\, f(t_1, \ldots, t_n) \,|_v^{\langle U, \varrho \rangle} = \varrho(f)(|\, t_1 \,|_v^{\langle U, \varrho \rangle}, \ldots, |\, t_n \,|_v^{\langle U, \varrho \rangle})$$

  where $t_i \in Term$ for all $i \in \{1, 2 \ldots, n\}$.

First-order logic – evaluation of atomic formulae

- If $p \in \mathcal{P}(0)$, then $\mid p \mid_v^{\langle U, \varrho \rangle} = \varrho(p)$
- If $t_1, t_2 \in Term$, then

$$\mid (t_1 = t_2) \mid_v^{\langle U, \varrho \rangle} = \begin{cases} 1, & \text{if } \mid t_1 \mid_v^{\langle U, \varrho \rangle} = \mid t_2 \mid_v^{\langle U, \varrho \rangle} \\ 0, & \text{otherwise.} \end{cases}$$

- If $P \in \mathcal{P}(n)$ for some $n > 0$, then

$$\mid P(t_1, \ldots, t_n) \mid_v^{\langle U, \varrho \rangle} = \varrho(P)(\mid t_1 \mid_v^{\langle U, \varrho \rangle}, \ldots, \mid t_n \mid_v^{\langle U, \varrho \rangle})$$

where $t_i \in Term$ for all $i \in \{1, 2 \ldots, n\}$.

First-order logic – evaluation of compound formulae (1)

Evaluation rules inherited from zero-order logic

- If $A \in Form$, then $|\neg A|_v^{\langle U,\varrho\rangle} = 1 - |A|_v^{\langle U,\varrho\rangle}$.
- If $A, B \in Form$, then

$$|(A \supset B)|_v^{\langle U,\varrho\rangle} = \begin{cases} 0 & \text{if } |A|_v^{\langle U,\varrho\rangle} = 1, \text{ and } |B|_v^{\langle U,\varrho\rangle} = 0; \\ 1, & \text{otherwise.} \end{cases}$$

$$|(A \wedge B)|_v^{\langle U,\varrho\rangle} = \begin{cases} 1 & \text{if } |A|_v^{\langle U,\varrho\rangle} = 1, \text{ and } |B|_v^{\langle U,\varrho\rangle} = 1; \\ 0, & \text{otherwise.} \end{cases}$$

$$|(A \vee B)|_v^{\langle U,\varrho\rangle} = \begin{cases} 0 & \text{if } |A|_v^{\langle U,\varrho\rangle} = 0, \text{ and } |B|_v^{\langle U,\varrho\rangle} = 0; \\ 1, & \text{otherwise.} \end{cases}$$

$$|(A \equiv B)|_v^{\langle U,\varrho\rangle} = \begin{cases} 1 & \text{if } |A|_v^{\langle U,\varrho\rangle} = |B|_v^{\langle U,\varrho\rangle} \\ 0, & \text{otherwise.} \end{cases}$$

First-order logic – evaluation of compound formulae (2)

Semantic rules for quantification

- If $A \in Form, x \in Var$, then

$$|\forall x A|_v^{\langle U, \varrho \rangle} = \begin{cases} 0, & \text{if there exists an } u \in U, \text{ such that } |A|_{v[x:u]}^{\langle U, \varrho \rangle} = 0; \\ 1, & \text{otherwise.} \end{cases}$$

$$|\exists x A|_v^{\langle U, \varrho \rangle} = \begin{cases} 1, & \text{if there exists an } u \in U, \text{ such that } |A|_{v[x:u]}^{\langle U, \varrho \rangle} = 1; \\ 0, & \text{otherwise.} \end{cases}$$